Hand-in date: see Canvas, individually.

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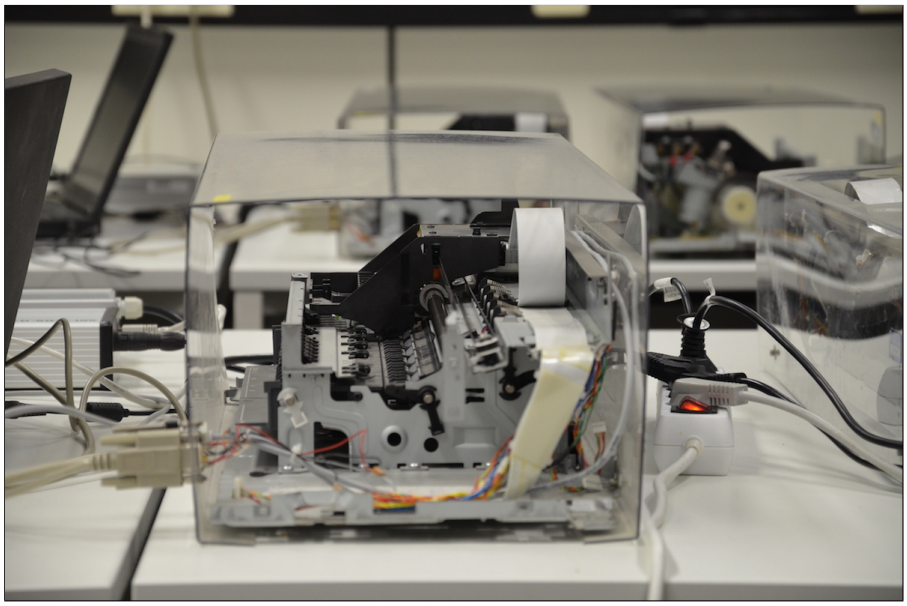
1. **Frequency domain ILC applied to a printer system**

   In this exercise, a frequency domain ILC controller is designed and implemented on a simulation model of a printer system, see the figure, below. Indeed, repeating tasks are standard in printing systems, see also [1].



Printer system.

1.a Load the frequency response function, the parametric model G and controller C. Inspect the results and inspect the following aspects: continuous/discrete time model? sampling frequency/time? Nyquist frequency? Bode diagram: collocated/non-collocated2 mass-spring-damper system? model order? model stable? what kind of controller? controller stable? controller order? Closed-loop stable? bandwidth, gain margin, phase margin, modulus margin?

1.b Perform a simulation with the feedback controller implemented. Set `N_trial = 1` and use `SimILC.m`. Inspect the servo error. What is the achieved accuracy?

1.c Implement a mass feedforward on the simulation model, i.e., $f = C^{\text{ff}}\ddot{r}$ Derive a suitable value of m based on G. Implement the result by implementing the feedforward controller in `FeedforwardUpdate.m`, set `N_trial = 2`, and using `SimILC.m`. What accuracy can you achieve?

1.d Next, we proceed with designing a frequency domain ILC controller. Construct L by using non-causal inversion techniques. It is recommended to use stable inversion as in the slides, which is only a few lines in Matlab. Alternatively, if you use ZPETC, make sure to construct both the causal L_c and the non-causal L, investigate the quality of the inverse using a Bode diagram, check How large are magnitude and phase errors for L and L_c, investigate how large is `p+d` and why, and finally the order of L_c?

1.e Investigate whether the designed L leads to a convergent ILC scheme when evaluated for the model GS. Next, investigate whether the designed learning filter L leads to a convergent ILC scheme when evaluated for the frequency response function of GS. (hint: use `bodemag.m`.)

1.f Design a robustness filter Q to ensure that the convergence condition is met for both the model and the frequency response function (hint: use `butter.m`). Notice that the frequency response function should be ignored below 20 Hz due to a poor signal to noise ratio. Generate relevant (Bode) plots and explicitly indicate which function you plot.

1.g Implement the controller in `FeedforwardUpdate.m`, set `N_trial` to your desired value, and use `SimILC.m` to run simulations and possibly `stableinv.m` or `zpetc.m`. Interpret the results. Do the results coincide with your expectations?

1.h Incorporate a learning gain α, i.e. replace L by αL. Explain the purpose of α. What is a suitable range for α given its purpose? What is the admissible range of α to ensure convergence (assuming $L = (GS)^{-1}$ and $Q = 1$)? Investigate the influence of the learning gain on the convergence speed using simulations.

2. **Frequency domain ILC applied to a printer system: Experiments**

2.a (**Experiment.**) Implement the obtained mass feedforward of Exercise c on the experimental setup. Implement the feedforward controller in `feedForwardUpdate.m`, set `N_trial = 2`, and using `ExpILC.m`. Compare the result without (trial 1) and with (trial 2) feedforward. What is the achieved accuracy? What is the performance increase/decrease? Compare this with the result of Exercise c and explain the observed differences. Try to tune the value of m further, what accuracy can you achieve?

2.b (**Experiment**) Implement the designed controller in Exercise g and h experimentally. Start with a small learning gain α. Interpret the results, is convergence achieved? What is the peak servo error and 2-norm of the servo error? Investigate the dependence of the results on Q and α. What is the best servo performance that can be achieved in terms of peak and 2-norm?

3. **Exercise Repetitive Control**

In this exercise, a repetitive controller is designed for the printer setup.

1. To implement a repetitive controller, a model of the system is required.
   (a) Which model is required to construct L? Construct the required model. What is the order? Please explain. (hint: use `feedback.m`)
   (b) Is the model stable? minimum phase? proper/nonproper/strictly proper? (hint: use `zpkdata.m`)
2. Next, the learning filter L and robustness filter Q are designed.
   (a) Construct L using `ZPETC.m`. Construct both the causal L_c and the non-causal L (seeILC part, section implementation). Investigate the quality of the inverse using a Bode diagram. How large are magnitude and phase errors for L and L_c? How large is `p+d` and why?

(b) Investigate whether the designed learning filter L leads to a stable RC scheme when evaluated for the model T. Next, investigate whether the designed learning filter L leads to a stable RC scheme when evaluated for the frequency response function of T. (hint: use `bodemag.m`. See also slide 16)

(c) Design a robustness filter Q using `fir1.m` to ensure that the stability condition (slide 17) is met for both the model and the frequency response function. Notice that the frequency response function should be ignored below 20 Hz due to a poor signal to noise ratio. Draw a Bode diagram of Q and Q^*Q. Explain magnitude and phase. Investigate the influence of the order of Q and the cut-off frequency. How large is `p_Q` and why?

3. The next steps involve simulations.

   (a) Save your controller, and open the Simulink model `PrinterRC_SIM.mdl`. Open the block 'repetitive controller' and interpret the block scheme (see also slide 16 and especially 19).

   (b) Open `SimExpRC.m`. Set $N_{\text{trial}} = 10$, $N_{\text{RCactivate}} = 5$, and method = 'simulation'. Run the file. Interpret the reference signal, and the simulation results. Do the results coincide with your expectations?

4. The next steps involve experiments.

   (a) (**Experiment**) In `SimExpRC.m`, set $N_{\text{trial}} = 20$, $N_{\text{RCactivate}} = 5$, and method = 'experiment'. Run the experiment, and interpret the results. What is the peak servo error, 2-norm of the servo error? What is the performance improvement compared to using only standard feedback control?

   (b) (**Optional**) Incorporate a learning gain α, i.e., replace L by αL. Explain the purpose of α. What is a suitable range for α given its purpose? What is the admissible range of α to ensure stability (assuming $L = T^{-1}$ and $Q = 1$)? Investigate the dependence of the results on L, Q, and α. What is the best servo performance that can be achieved?

4. **Lifted ILC applied to the printer system**

In this exercise, lifted ILC is applied to the printer system.

4.a For lifted ILC, a model of the system is required in the lifted domain. In particular, the matrix J will contain the impulse response of GS. Set the trial length N equal to 100 and construct J (hint: use `dimpulse` and `toeplitz`. In addition, set small elements of J equal to zero to facilitate the later computations). Inspect its structure (hint: use `mesh` and `imagesc`). Is the matrix triangular? Toeplitz? Is the diagonal equal to zero? How does this relate to system properties, e.g., causality, time (in)variance, properness?

4.b Next, a lifted ILC controller will be designed. Keep $N = 100$, to facilitate fast computations. Set $W_e = w_e I$, $W_f = w_f I$, and $W_{\Delta f} = w_{\Delta f} I$, and choose as initial values, e.g., $w_e = 10^3$, $w_f = 10^{-3}$, $w_{\Delta f} = 10^{-1}$. Compute the optimal lifted ILC controller L and Q. Inspect the resulting matrices L and Q (hint: use `plot(diag(X,i))`) to plot the ith diagonal of the matrix X (use negative i to obtain lower diagonals))). What is the structure? How does this relate to causality and time (in)variance? Vary the values of w_e, w_f, $w_{\Delta f}$. What happens to L and Q if w_f is set to zero? How should the values be chosen to guarantee $e_\infty = 0$?

4.c Investigate convergence and monotonic convergence properties of the lifted ILC controller.

4.d Set $N = 4501$, i.e., the length of the servo task. Implement and simulate the lifted ILC controller. Investigate the influence of w_e, w_f, $w_{\Delta f}$ on convergence speed and ILC performance.

4.e (**Experiment**) This exercise has been removed.

4.f Compare the lifted ILC controller to the frequency domain ILC controller. Comment on

(a) how to design for zero converged error,
(b) how to attenuate trial varying disturbances,
(c) how to address model errors,
(d) computational speed for the used implementation (note that there are more efficient implementations of lifted ILC, see lecture slides), and
(e) causality and time-(in)variance of the ILC controller.

5. Consider the system

$$J = \frac{1}{z} \tag{1}$$

and the ILC controller $f_{j+1} = Qf_j + Le_j$, with

$$Q = 1 \tag{2}$$
$$L = \alpha z, \quad \alpha \in \mathbb{R}. \tag{3}$$

1. Consider an infinite time interval, i.e., frequency domain ILC. For which values of α is the system monotonically convergent? Why?
2. Next, the ILC controller (2)-(3) is implemented in finite time with $N = 2$. What are the corresponding Toeplitz matrices? Is the system (monotonically) convergent?

6. Let $J = 1$ with trial length $N = 1$. Assume the ILC controller $f_{j+1} = Qf_j + Le_j$ with $Q = -0.8$, $L = -0.4$. Set the reference $r = 1$ and perform 20 iterations. Plot e_k and f_k. What can you conclude from the plots regarding convergence? Monotonic convergence?

7. Consider the system

$$J(z) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] = \left[\begin{array}{c|c} 0.5 & 1 \\ \hline 1 & 1 \end{array}\right] \tag{4}$$

and the ILC controller $f_{j+1} = Qf_j + Le_j$, with

$$Q = 1 \tag{5}$$
$$L = \alpha, \quad \alpha \in \mathbb{R}. \tag{6}$$

In terms of the underlying feedback loop, the closed-loop system corresponds to $J = SP$ and $C = 1$.

1. Consider an infinite time interval, i.e., frequency-domain ILC. For which values of α is the system convergent?

2. Next, the ILC scheme (5), (6) is implemented in finite time with $N = 2$. What is the admissible range of α for convergence? And for monotonic convergence? Note: to analyze monotonic convergence, you are allowed to pursue a numerical approach by gridding α.
3. What is the admissible range of α to guarantee convergence (not necessarily monotonic) on an arbitrary finite time interval $N \in \mathbb{N}$?
4. Suppose your colleague designed a lifted ILC $L = 0.75 \cdot I \in \mathbb{R}^{2\times 2}$, i.e., for trial length $N = 2$. However, the ILC was implemented in an RC setting, i.e., the states of $J(z)$ are *not* reset after every trial, and the system is excited by a periodic $r(k)$, where $r(k+N) = r(k)$.

   Can you derive a condition for stability of the closed-loop system? Is the condition sufficient? Is the condition necessary? Apply the obtained condition, can you conclude whether the system is stable?
5. Same question as (d), but now for $N = 3$.

References

[1] J. Bolder, T. Oomen, S. Koekebakker, and M. Steinbuch. Using iterative learning control with basis functions to compensate medium deformation in a wide-format inkjet printer. *Mechatronics*, 24(8):944–953, 2014.